Functional Annotation Report: UbiH (Q88CI2 / PP_5199) in *Pseudomonas putida* KT2440

Summary

UbiH (UniProt Q88CI2; locus PP_5199) of *Pseudomonas putida* KT2440 is a flavin (FAD)-dependent, NAD(P)H- and O_2-consuming aromatic-ring monooxygenase that performs one of the three oxygenative hydroxylation steps in the biosynthesis of coenzyme Q (ubiquinone). Its primary reaction is the *para* (C4) hydroxylation of the lipophilic quinone precursor, converting **2-polyprenyl-6-methoxyphenol → 2-polyprenyl-6-methoxy-1,4-benzoquinol**. Because *P. putida* synthesizes ubiquinone-9 (Q9), the physiological substrate carries a nonaprenyl tail rather than the octaprenyl tail found in *E. coli* (Q8) — hence the historical name "2-octaprenyl-6-methoxyphenyl hydroxylase." The enzyme uses a non-covalently bound FAD cofactor, molecular O_2, and NAD(P)H as reductant.

The functional assignment is secure. The gene is annotated as *ubiH* in UniProt, KEGG (ppu:PP_5199, orthology K03185, EC 1.14.13.-), and InterPro, the protein belongs to the UbiH/COQ6 monooxygenase family, and the *P. putida* protein is a genuine ortholog (51% amino-acid identity) of the biochemically characterized *E. coli* UbiH. The enzyme adopts the canonical **p-hydroxybenzoate hydroxylase (PHBH) fold** of single-component "class A" flavoprotein aromatic hydroxylases; its N-terminal region contains the diagnostic FAD/NAD(P)-binding Rossmann dinucleotide motif (GxGxxG, present as **GGGLVG** at residues 10–15 of Q88CI2).

Localization and pathway context: most ubiquinone-biosynthesis steps occur in the **bacterial cytosol**, where the Ubi enzymes assemble into a soluble multiprotein complex — the "Ubi metabolon" — that relies on lipid-binding accessory proteins (UbiJ/UbiK) to sequester the extremely hydrophobic polyprenyl-tailed intermediates. UbiH therefore acts as a soluble/peripheral cytoplasmic enzyme carrying out its chemistry on a membrane-partitioned lipid substrate at the cytoplasmic face of the inner membrane. Its product feeds forward toward mature coenzyme Q, the lipophilic electron/proton carrier of the aerobic respiratory chain.

Consistent with a purely aerobic role, *ubiH*-type mutants are blocked in the O_2-dependent hydroxylation branch but can still make ubiquinone anaerobically via O_2-independent alternative hydroxylases.

Key Findings

Finding 1 — UbiH is the C4 (*para*) aromatic-ring hydroxylase of ubiquinone biosynthesis

UniProt Q88CI2 is annotated as "2-octaprenyl-6-methoxyphenyl hydroxylase," gene *ubiH*, in *P. putida* KT2440, and belongs to the UbiH/COQ6 family (InterPro IPR011295 "UbiH"; IPR010971 "UbiH/COQ6"). The functional assignment rests on the well-characterized *E. coli* ortholog. In *E. coli*, three of the ubiquinone-pathway reactions are hydroxylations that introduce hydroxyl groups at contiguous positions of the benzene nucleus, catalyzed by three flavin monooxygenases; **UbiH performs the C-4 (*para*) hydroxylation** (PMID: 10802164). The biosynthesis of ubiquinone requires exactly three hydroxylation reactions on contiguous positions of an aromatic ring (PMID: 27822549).

The specific transformation is **2-polyprenyl-6-methoxyphenol → 2-polyprenyl-6-methoxy-1,4-benzoquinol**: introduction of the *para* hydroxyl converts the mono-phenol into a 1,4-dihydroxy (hydroquinone) ring. In *E. coli* the substrate is the octaprenyl homolog; in *P. putida*, which produces Q9, the substrate carries a nonaprenyl tail. Substrate specificity is defined by (i) the methoxy/prenyl-substituted phenolic ring recognized in the active site and (ii) the long polyisoprenoid tail, which is accommodated and sequestered by the pathway's lipid-binding accessory proteins rather than fully buried in the monooxygenase. This answers the central "what reaction is catalyzed" question: an FAD-dependent, regioselective *para*-hydroxylation of a polyprenylated methoxyphenol intermediate.

> *"Three of these reactions involve hydroxylations resulting in the introduction of hydroxyl groups at positions C-6, C-4, and C-5 of the benzene nucleus of Q. The genes encoding the enzymes responsible for these hydroxylations, ubiB, ubiH, and ubiF"* — PMID: 10802164
>
> *"The biosynthesis of UQ requires three hydroxylation reactions on contiguous positions of an aromatic ring."* — PMID: 27822549

Finding 2 — UbiH is an FAD-dependent (class A) monooxygenase of the PHBH structural family, using O_2 and NAD(P)H

Q88CI2 carries FAD-binding domains (InterPro IPR002938 "FAD-bd"; IPR036188 "FAD/NAD(P)-binding superfamily") and is a member of the UbiH/COQ6 monooxygenase family (IPR051205). The structural template is established: the related Q-biosynthesis monooxygenase **UbiI was crystallized and shown to share the canonical FAD-dependent p-hydroxybenzoate hydroxylase (PHBH) fold**, with a flavin-binding pocket essential for activity — the first structural characterization of a monooxygenase in Q biosynthesis (PMID: 23709220).

Mechanistically, such enzymes reduce the bound FAD with NAD(P)H, react the reduced flavin with molecular O_2 to form a **C4a-(hydro)peroxyflavin** intermediate, and use this reactive oxygen species for electrophilic aromatic hydroxylation — explaining the enzyme's strict **oxygen dependence**. One atom of O_2 is incorporated into the product hydroxyl; the other is reduced to water. This aerobic dependence is demonstrated genetically: mutants blocked in any of the three aerobic hydroxylation reactions (*ubiB*, *ubiH*, or *ubiF*) can still synthesize ubiquinone anaerobically, because O_2-independent alternative hydroxylases substitute for them under anoxic conditions (PMID: 365223). This places UbiH firmly in the O_2-dependent (aerobic) branch and identifies its co-substrates as molecular oxygen and NAD(P)H.

> *"This structure shares many features with the canonical FAD-dependent para-hydroxybenzoate hydroxylase and represents the first structural characterization of a monooxygenase involved in Q biosynthesis."* — PMID: 23709220
>
> *"mutants blocked in either of the three hydroxylation reactions of the aerobic pathway (ubiB, ubiH, or ubiF) are each able to synthesize ubiquinone 8 anaerobically"* — PMID: 365223

Finding 3 — Sequence-level evidence: Q88CI2 carries the N-terminal FAD-binding dinucleotide motif of class A flavoprotein hydroxylases

The UniProt Q88CI2 sequence (399 aa; header confirms "2-octaprenyl-6-methoxyphenyl hydroxylase … GN=ubiH … *Pseudomonas putida* KT2440") begins **MSRVNLAIIGGGLVGASLALALQAGA…**, containing the canonical FAD/NAD(P)-binding Rossmann dinucleotide-binding motif **GxGxxG** (present as **GGGLVG** at residues 10–15). This βαβ fingerprint is the ADP-of-FAD binding signature shared by the PHBH family of single-component FAD-dependent aromatic monooxygenases. It is fully consistent with the InterPro annotations IPR002938 (FAD-binding), IPR036188 (FAD/NAD(P)-binding superfamily), IPR011295 (UbiH), IPR010971 (UbiH/COQ6), and IPR051205 (UbiH/COQ6 monooxygenase). The direct presence of this motif in the target sequence provides independent, bioinformatic corroboration of the cofactor requirement inferred from family membership.

Finding 4 — Q88CI2 is a bona fide ortholog of biochemically characterized *E. coli* UbiH (51% identity)

A global (Needleman–Wunsch) alignment of the full-length target Q88CI2 (399 aa) against *E. coli* UBIH_ECOLI (P25534, 392 aa, "2-octaprenyl-6-methoxyphenol hydroxylase") gives **185/362 = 51.1% amino-acid identity** over the aligned length. The same target aligns to the family paralog UbiI (P25535, "2-octaprenylphenol hydroxylase") at only **165/364 = 45.3% identity**.

Comparison	Identity	Aligned length	Interpretation
Q88CI2 vs *E. coli* **UbiH** (P25534)	**51.1%**	362 aa	True ortholog — direct functional transfer justified
Q88CI2 vs *E. coli* **UbiI** (P25535)	45.3%	364 aa	Paralog — same family, different regioselectivity

The higher similarity to UbiH than to the paralogous ring-hydroxylase UbiI, combined with >50% full-length identity (well above the ~25–30% "twilight zone" of ambiguous homology), indicates **true orthology rather than mere family membership**. This is the strongest single piece of evidence permitting direct transfer of the experimentally established *E. coli* UbiH function (C4 *para*-hydroxylation) to the *P. putida* protein.

Finding 5 — Conservation is uniform across FAD-binding and substrate-binding regions, supporting conserved substrate specificity

Regional identity from the global pairwise alignment of Q88CI2 vs *E. coli* UbiH (P25534):

Region	Approx. residues	Role in PHBH-family enzymes	Identity
N-terminal third	~1–133	FAD dinucleotide binding	50.4%
Middle third	~134–266	Core / linker	49.2%
C-terminal third	~267–399	Aromatic-substrate pocket / dimerization	**53.8%**

Conservation is thus **not restricted to the cofactor-binding module** but is in fact slightly *higher* in the C-terminal region that, in PHBH-family monooxygenases, forms the aromatic-substrate binding pocket. Because the substrate-recognition machinery is retained at least as strongly as the cofactor machinery, the *P. putida* enzyme is predicted to have the **same substrate specificity and regioselectivity** as *E. coli* UbiH — the C4 hydroxylation of 2-polyprenyl-6-methoxyphenol, not merely generic hydroxylase chemistry. The diagnostic N-terminal GxGxxG (GGGLVG) FAD motif is present as expected.

Finding 6 — UbiH acts in the cytosolic Ubi metabolon on a membrane-embedded lipophilic substrate

Most steps of ubiquinone biosynthesis take place in the **bacterial cytosol**, where the Ubi enzymes assemble into a soluble multiprotein complex — the "Ubi metabolon." This assembly requires the accessory proteins **UbiJ** (bearing an SCP2 lipid-binding domain) and **UbiK** to sequester the polyprenyl-tailed intermediates, which are otherwise too hydrophobic to remain in solution (PMID: 36142227; PMID: 38710096). UbiH is therefore a **soluble/peripheral cytoplasmic enzyme** (not an integral membrane protein) that carries out its chemistry on a membrane-partitioned lipid substrate at the **cytoplasmic face of the inner membrane** within this complex.

In flux terms, UbiH functions as a downstream-pulling step: in *Rhodobacter sphaeroides*, co-overexpression of UbiH relieved accumulation of an upstream intermediate and increased CoQ10 yield, demonstrating that UbiH activity pulls metabolic flux toward downstream CoQ intermediates (PMID: 25817210). This answers the "where does the gene product act" question: it is a soluble, cytosolic enzyme operating on a membrane-partitioned lipid at the inner-membrane interface as part of a defined biosynthetic complex, and its end product, coenzyme Q, is the central lipophilic electron/proton carrier of the aerobic respiratory chain.

> *"Most steps of UQ biosynthesis take place in the cytosol of..."* — PMID: 36142227
>
> *"UbiH was then co-overexpressed to pull the metabolic flux towards downstream..."* — PMID: 25817210

Finding 7 — KEGG confirms the assignment and places PP_5199 in a complete *P. putida* ubiquinone pathway

The KEGG entry for **ppu:PP_5199** assigns ORTHOLOGY **K03185** "2-octaprenyl-6-methoxyphenol hydroxylase [EC:1.14.13.-]"; PATHWAY **ppu00130** (Ubiquinone and other terpenoid-quinone biosynthesis); MODULE **ppu_M00117** (Ubiquinone biosynthesis, prokaryotes: chorismate + polyprenyl-PP ⇒ ubiquinol); and MOTIF Pfam **FAD_binding_3 (PF01494)** — the PHBH-family FAD-binding domain. The EC class **1.14.13.-** denotes an NAD(P)H- and O_2-dependent monooxygenase, consistent with the mechanism in Finding 2.

Genomic-neighborhood analysis (POSITION complement 5930870..5932069) shows PP_5199 (*ubiH*) flanked by **PP_5197** (K18800/*ubiV*, "oxidoreductase involved in anaerobic synthesis of ubiquinone") and PP_5198 (a small conserved protein of unknown function), with *pepP* (PP_5200) beyond. This adjacency of the aerobic UbiH to the O_2-independent (anaerobic) hydroxylase *ubiV* mirrors the two-branch aerobic/anaerobic ring-hydroxylation system. Importantly, *P. putida* also encodes a **complete Ubi pathway** elsewhere in the genome: *ubiA* (PP_5318), *ubiD/ubiX* (PP_5213/PP_5317), *ubiE* (PP_5011), *ubiC* (PP_5013), plus additional UbiF-type monooxygenases (PP_0427, PP_0548). The presence of a full, functional Ubi gene set confirms that PP_5199 is a committed, non-redundant step in an active coenzyme-Q biosynthetic route, not an orphan gene.

Mechanistic Model / Interpretation

Position of UbiH in the ubiquinone biosynthetic pathway

Ubiquinone biosynthesis in proteobacteria proceeds from **chorismate** (via 4-hydroxybenzoate, 4-HB) and a **polyprenyl-diphosphate** tail (nonaprenyl-PP in *P. putida*). The aromatic ring is prenylated (UbiA), decarboxylated (UbiD/UbiX), and then subjected to alternating **hydroxylation** (Ubi monooxygenases) and **O-/C-methylation** (UbiG/UbiE) steps that decorate the ring into the final ubiquinone quinone head group. UbiH performs the **C4 (*para*) hydroxylation**, one of the three contiguous ring hydroxylations:

```
Chorismate ──> 4-hydroxybenzoate (4-HB)
                    │  UbiA (prenylation, adds C45 nonaprenyl tail)
                    ▼
        3-nonaprenyl-4-hydroxybenzoate
                    │  UbiD / UbiX (decarboxylation)
                    ▼
             2-nonaprenylphenol
                    │  UbiI-type (ring hydroxylation, O2)
                    ▼
        2-nonaprenyl-6-hydroxyphenol
                    │  UbiG (O-methylation)
                    ▼
        2-nonaprenyl-6-methoxyphenol   ◄── UbiH SUBSTRATE
                    │  ****  UbiH  ****  (C4 / para hydroxylation; O2 + NAD(P)H; FAD)
                    ▼
   2-nonaprenyl-6-methoxy-1,4-benzoquinol   ◄── UbiH PRODUCT
                    │  UbiE (C-methylation)
                    ▼
                    │  UbiF-type (ring hydroxylation, O2)  +  UbiG (O-methylation)
                    ▼
              Ubiquinol-9 (reduced Q9)  ⇌  Ubiquinone-9 (Q9)
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(Intermediate order follows the E. coli paradigm; the P. putida homologs carry a nonaprenyl rather than octaprenyl tail because the organism makes Q9. Regioselectivity nomenclature (C1/C4/C5/C6) varies between older and revised models, but the transformation mono-phenol → 1,4-benzoquinol is unambiguous.)

Catalytic cycle (class A flavoprotein monooxygenase / PHBH paradigm)

```
   NAD(P)H ──► reduces bound FAD ──► FADH⁻
                                      │
                                      │ + O2
                                      ▼
                          C4a-(hydro)peroxyflavin
                                      │  electrophilic aromatic
                                      │  hydroxylation at ring C4
                                      ▼
     product-OH  +  C4a-hydroxyflavin ──► dehydration ──► oxidized FAD + H2O
```

Localization and supramolecular organization

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        CYTOPLASM
    ┌────────────────────────────────────────────────┐
    │   Ubi metabolon (soluble multiprotein complex) │
    │   UbiE UbiF UbiG UbiH UbiI ...                  │
    │        │      + UbiJ (SCP2 lipid carrier)       │
    │        │      + UbiK                            │
    │        ▼                                        │
    │   UbiH acts on lipid substrate held by UbiJ    │
    └────────┼───────────────────────────────────────┘
     ~~~~~~~~▼~~~~~~~ cytoplasmic face ~~~~~~~~~~~~~~~~
    ████████████ INNER MEMBRANE ████████████████████
         (polyprenyl-tailed intermediate partitions here;
          mature Q9 becomes the respiratory-chain quinone)
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Summary annotation table

Attribute	Assignment	Confidence
Primary function	FAD-dependent aromatic *para* (C4) monooxygenase	High
Reaction	2-nonaprenyl-6-methoxyphenol → 2-nonaprenyl-6-methoxy-1,4-benzoquinol	High
Co-substrates / cofactor	O_2, NAD(P)H; non-covalent FAD	High
EC number	1.14.13.-	High (KEGG K03185)
Pathway	Ubiquinone (coenzyme Q9) biosynthesis, aerobic branch	High
Localization	Cytosol / cytoplasmic face of inner membrane (Ubi metabolon)	Medium–High
Substrate specificity	Polyprenyl-methoxyphenol; conserved with *E. coli* UbiH	Medium–High (inferred from orthology)
Structural fold	p-hydroxybenzoate hydroxylase (PHBH) fold	High (by homology)

Evidence Base

PMID	Title (abbrev.)	How it supports the annotation
10802164	*Ubiquinone biosynthesis in E. coli: identification of ubiF*	Assigns UbiH to the C-4 (*para*) aromatic-ring hydroxylation; names the three ring hydroxylases of the Q pathway (Finding 1).
27822549	*Evolution of Ubiquinone Biosynthesis…*	Establishes that UQ biosynthesis needs three hydroxylations on contiguous ring positions; frames family regioselectivity (Findings 1, 5).
23709220	*ubiI, a new gene in E. coli CoQ biosynthesis…*	First crystal structure of a Q-pathway monooxygenase (UbiI); shows the PHBH FAD-dependent fold shared by UbiH-family enzymes (Finding 2).
365223	*Alternative hydroxylases for aerobic/anaerobic UQ biosynthesis*	Demonstrates UbiH functions in the O_2-dependent aerobic branch; anaerobic alternatives replace it without O_2 (Finding 2).
36142227	*Functional Role of UbiJ–UbiK…*	Localizes the Ubi enzymes to the cytosol and defines the soluble Ubi metabolon requiring lipid-binding accessory proteins (Finding 6).
38710096	*Structural Reconstruction of the Ubi complex*	Supports the metabolon model and sequestration of hydrophobic intermediates (Finding 6).
25817210	*Metabolic bottlenecks for CoQ10 in R. sphaeroides*	Shows UbiH overexpression pulls flux toward downstream CoQ intermediates, confirming its on-pathway position (Finding 6).
7032590	*Respiratory mutant of E. coli with reduced aminoglycoside uptake*	Illustrates the respiratory consequences of defective ubiquinone ring modification (physiological context).

Supplementary database evidence (KEGG, UniProt, InterPro, Pfam) underpins Findings 3, 4, 5, and 7: the KEGG ortholog assignment K03185/EC 1.14.13.-, the InterPro/Pfam FAD-binding and UbiH/COQ6 domain annotations, the presence of the GGGLVG FAD motif in the target sequence, and the 51% ortholog-level identity to *E. coli* UbiH derived from pairwise global alignment.

Supported and Refuted Hypotheses

Supported

- **H1** — UbiH is an enzyme (monooxygenase), not a transporter/structural protein — *supported* (family, domains, sequence motif, biochemical literature).
- **H2** — UbiH catalyzes a specific aromatic C4/*para* hydroxylation in Q biosynthesis — *supported* (PMID: 10802164; PMID: 27822549).
- **H3** — UbiH is FAD-dependent, O_2- and NAD(P)H-utilizing, PHBH-fold — *supported* (PMID: 23709220; sequence Rossmann motif; anaerobic bypass in PMID: 365223).
- **H4** — UbiH functions in the cytosol/Ubi metabolon on a lipid substrate — *supported* (PMID: 36142227; PMID: 38710096).

Refuted / ruled out

- UbiH is **not** an oxygen-independent enzyme — anaerobic Q synthesis proceeds via distinct alternative hydroxylases (PMID: 365223).
- UbiH is **not** an integral inner-membrane transporter or a menaquinone-pathway enzyme; it is a cytosolic flavoprotein specific to the ubiquinone ring-modification steps.

Limitations and Knowledge Gaps

1. **No direct biochemical characterization of the *P. putida* protein.** The functional assignment for Q88CI2 rests on orthology transfer (51% identity to *E. coli* UbiH) and domain/family membership rather than on purified-enzyme assays of PP_5199 itself. No published kinetic constants (Km, kcat), substrate-range data, or in-vitro reconstitution exist specifically for the *P. putida* enzyme.
2. **No experimental structure.** The PHBH-fold assignment is inferred from the UbiI crystal structure and sequence homology; there is no experimental 3D structure of Q88CI2 confirming the flavin pocket and substrate channel.
3. **Substrate tail length inferred, not measured.** The nonaprenyl (Q9) substrate is inferred from *P. putida*'s known quinone content; the exact intermediate handled by PP_5199 has not been directly identified in this strain.

4. **Regioselectivity edge cases.** The UbiH/UbiF/UbiI family shows documented plasticity in regioselectivity across proteobacteria (PMID: 27822549). While the high C-terminal conservation argues for a conserved C4 specificity, cross-complementation has not been demonstrated for the *P. putida* protein specifically. Regioselectivity nomenclature (C1/C4 conventions) also varies between older and revised models.
5. **Localization details.** The cytosolic/metabolon localization is established for *E. coli*; whether *P. putida* assembles an identical UbiJ/UbiK-dependent complex is assumed by homology, not directly shown. *P. putida* UbiK/UbiJ orthology was not separately verified.
6. **Genomic-context nuance.** PP_5199 lies adjacent to *ubiV* (anaerobic UQ synthesis) rather than in an operon with the canonical aerobic *ubi* genes, which are dispersed across the genome. The regulatory logic of this arrangement was not explored.

Proposed Follow-up Experiments / Actions

1. **Genetic complementation.** Test whether cloned *P. putida* PP_5199 restores ubiquinone synthesis and aerobic respiratory growth in an *E. coli ΔubiH* mutant (succinate-oxidation / growth-on-nonfermentable-carbon phenotype). This would directly confirm functional equivalence.
2. **In-vitro enzymology.** Heterologously express and purify Q88CI2, reconstitute with FAD, and assay *para*-hydroxylation of a synthetic 2-prenyl-6-methoxyphenol analog in the presence of O_2 and NAD(P)H. Measure Km/kcat and confirm C4 regioselectivity by MS/NMR of the product.
3. **Structural determination or high-confidence modeling.** Obtain an AlphaFold model (or crystal structure) of Q88CI2, verify the PHBH fold, dock the polyprenyl-methoxyphenol substrate, and identify active-site residues; compare with the UbiI structure (PMID: 23709220).
4. **Knockout phenotyping in *P. putida* KT2440.** Delete PP_5199 and quantify ubiquinone-9 levels, accumulation of the expected upstream intermediate (2-nonaprenyl-6-methoxyphenol), and aerobic vs. anaerobic respiratory competence.
5. **Metabolon verification.** Test for co-assembly of PP_5199 with *P. putida* UbiJ/UbiK orthologs (co-purification / crosslinking) to confirm the cytosolic-metabolon localization model in this organism.

6. **Regioselectivity mapping.** Given family plasticity, perform structure-guided mutagenesis of the C-terminal substrate-pocket residues to test whether specificity/regioselectivity can be switched, informing the evolutionary framework of PMID: 27822549.

Gene-Identity Verification Statement

The gene symbol **ubiH** matches the UniProt protein description ("2-octaprenyl-6-methoxyphenyl hydroxylase"), the organism (*Pseudomonas putida* KT2440), the locus (PP_5199), the family (UbiH/COQ6), and the FAD-binding domain architecture. These are consistent across UniProt, KEGG (ppu:PP_5199, K03185), and the sequence header. The literature retrieved concerns the same UbiH/coenzyme-Q system across proteobacteria; **no evidence of an unrelated same-symbol gene was found.** Function was transferred to the *P. putida* protein on the basis of demonstrated orthology (51% identity to biochemically characterized *E. coli* UbiH), not merely a shared symbol. The annotation is therefore made on the correct protein and is judged secure.

Cited References

- Kwon, Kotsakis & Meganathan (2000) *J Bacteriol.* — PMID: 10802164
- Pelosi et al. (2016) *mBio.* — PMID: 27822549
- Hajj Chehade et al. (2013) *J Biol Chem.* — PMID: 23709220
- Alexander & Young (1978) *J Bacteriol.* — PMID: 365223
- Lu et al. (2015) *Metab Eng.* — PMID: 25817210
- Launay et al. (2022) — PMID: 36142227; Launay et al. (2024) — PMID: 38710096
- Muir, Hanwell & Wallace (1981) — PMID: 7032590
